



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2005

YEAR 12

ASSESSMENT TASK #3

Mathematics Extension 1

General Instructions

- Working time – 90 minutes.
- Reading Time – 5 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may not be awarded for messy or badly arranged work
- Hand in your answer booklets in 3 sections. Section A (Question 1), Section B (Question 2) and Section C (Question 3)

Total Marks - 80

- Attempt questions 1 – 3
- All sections are NOT of equal value.

Examiner: *A. Fuller*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate

Total marks - 80

Attempt Questions 1 - 3

All questions are NOT of equal value

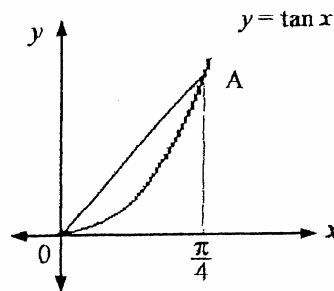
Answer each SECTION in a SEPARATE writing booklet.

Section A		Marks
Question 1 (27 marks)		
(a)	Evaluate $\log_2 0.125$	1
(b)	Expand $\left(e^{\frac{1}{2}x} - e^{-\frac{1}{2}x}\right)^2$	2
(c)	Evaluate $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$	2
(d)	Sketch the graph of $y = 3 \sin^{-1} \frac{x}{2}$	2
(e)	Prove that $\log_{ab} x = \frac{\log_a x}{1 + \log_a b}$	2
(f)	If $\int_0^1 \frac{1}{3+x^2} dx = a\pi$, find the value of a	2
(g)	Differentiate $\log_e(\sin^3 x)$ writing your answer in the simplest form	2
(h)	(I) For what values of x is $\sin^{-1} x$ defined?	1
	(II) Find the maximum value of $2x(1-x)$	1
	(III) Find the range of the function given by $f(x) = \sin^{-1}[2x(1-x)]$	2

(i) Use the substitution $u = x^2$ to find $\int_0^{\frac{1}{\sqrt{2}}} \frac{x}{\sqrt{1-x^4}} dx$ 3

(j) If $y = x^n e^{ax}$, show that $\frac{dy}{dx} - ay = \frac{ny}{x}$ 3

(k)



In the diagram, A is a point on the curve $y = \tan x$ with an x coordinate of $\frac{\pi}{4}$. The chord OA has been drawn from the origin to the point A.

Show that the area enclosed by the chord OA and the curve $y = \tan x$ between $x = 0$ and $x = \frac{\pi}{4}$ has a magnitude of $\frac{1}{8}(\pi - 4\ln 2)$ units² 4

End of Section A

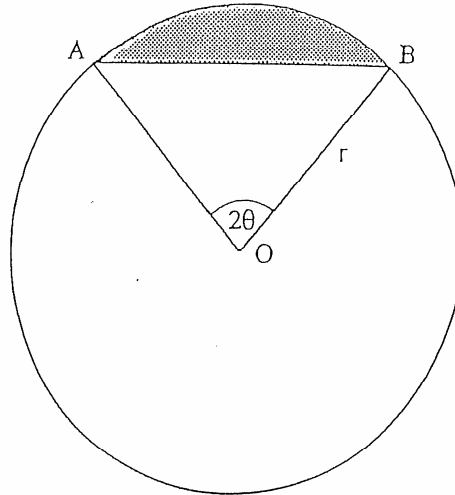
Section B (Use a SEPARATE writing booklet)

Marks

Question 2 (27 marks)

- (a) If $y = \sec x$, prove $\frac{dy}{dx} = \sec x \tan x$ **2**
- (b) A function is defined as $f(x) = 1 + e^{2x}$
- (I) Write down the domain and the range of the function **1**
- (II) Show that the inverse function can be defined **2**
as $f^{-1}(x) = \frac{1}{2} \ln(x-1)$
- (III) On the same set of axes, sketch the graphs of $y = f(x)$ **2**
and $y = f^{-1}(x)$
- (IV) Show that the equation of the normal to $y = f^{-1}(x)$ **2**
at the point where $f^{-1}(x) = 0$ is $2x + y - 4 = 0$
- (c) If $\frac{dy}{dx} = 1 + y$ and when $x = 0$, $y = 2$. Show that $y = 3e^x - 1$ **3**
- (d) Find the exact value of $\cos\left[2\sin^{-1}\frac{3}{4}\right]$ **3**
- (e) Evaluate $\int \frac{3x}{\sqrt{1+x}} dx$ using the substitution $u = 1 + x$ **3**
- (f) If $f(x) = a \cos^{-1}(bx)$, evaluate a and b if $f(0) = 2$ **3**
and $f'(0) = 2$.

(g)



The diagram above shows a shaded segment which subtends an angle of 2θ radians at the centre O of a circle with radius r . Given that the perimeter of the shaded segment equals twice the diameter of the circle

- | | | |
|-------|--|---|
| (I) | Show that $\sin\theta = 2 - \theta$ | 2 |
| (II) | Show that the equation $\sin\theta + \theta - 2 = 0$ has a root that lies between $\theta = 1$ and $\theta = 1.5$ | 1 |
| (III) | Use one application of Newton's method with an initial approximation of $\theta = 1.25$ to obtain a better approximation of the root of the equation $\sin\theta + \theta - 2 = 0$ | 2 |
| (IV) | Using the result found in (III) find to the nearest degree the size of $\angle AOB$ | 1 |

End of Section B

Section C (Use a SEPARATE writing booklet)

Marks

Question 3 (26 marks)

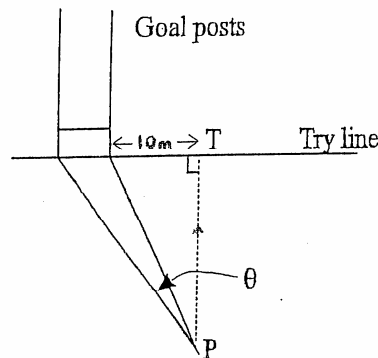
- (a) The velocity v of a particle moving along the x axis starting from $x = 1.8$ is given by $v = e^{-2x} \sqrt{2x^2 - 6}$, $x \geq 1.8$ where x is the displacement of the particle from the origin.
- (I) Show that the acceleration a of the particle in terms of its displacement can be expressed by $a = -2e^{-4x}(2x^2 - x - 6)$ 2
- (II) Hence, find the displacement of the particle at which the maximum speed occurs. 1
- (III) Show that the time T in seconds taken by the particle to move from $x = 2$ to $x = 3$ can be expressed as 2
- $$T = \int_2^3 \frac{e^{2x}}{\sqrt{2x^2 - 6}} dx$$
- (IV) Use Simpson's rule with three function values to obtain an approximate value for T . 2
- (b) A water tank is generated by rotating the curve $y = \frac{x^4}{16}$ around the y -axis.
- (I) Show that the volume of water, V as a function of its depth h , is given by $V = \frac{8}{3}\pi h^{\frac{3}{2}}$ 2
- (II) Water drains from the tank through a small hole at the bottom. The rate of change of the volume of water in the tank is proportional to the square root of the water's depth. 3
- Use this fact to show that the water level in the tank falls at a constant rate.

(c) Given $y = \cos^{-1}(\sin x)$

(I) Show that $\frac{dy}{dx}$ has two values 3

(II) Hence, or otherwise sketch the graph of $y = \cos^{-1}(\sin x)$ for $-\pi \leq x \leq \pi$ 2

(e) In rugby league, teams score points by placing the ball on (or over) the try line at the end of the field. A kicker may then convert the try by taking the ball back at right angles from the point T on the try line where the try was scored and attempt to kick the ball between the goal posts which are 6 metres apart.



In the diagram above, a try has been scored 10 metres to the right of the goal posts. The kicker has brought the ball back x metres to a point P to attempt the conversion. The kicker wants to maximise θ , the angle of his view of the goal posts.

(I) Show that $\tan \theta = \frac{6x}{160 + x^2}$. 3

(II) Letting $E = \tan \theta$, find the value of x for which E is a maximum. 2

(III) Hence show that the maximum angle, θ , is given by 2

$$\theta = \tan^{-1}\left(\frac{3}{\sqrt{160}}\right)$$

Question (e) continued over the page

- (IV) Find the maximum value of θ (to the nearest minute) and the corresponding value of x (to the nearest centimetre).

2

End of paper



SYDNEY BOYS HIGH SCHOOL
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2005
HIGHER SCHOOL CERTIFICATE
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Mathematics Extension 1

Sample Solutions

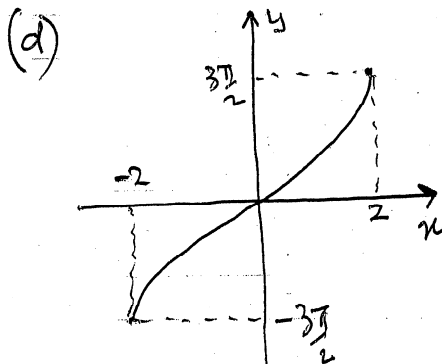
Section	Marker
A	AMG
B	FN
C	EC

Question 1

$$\begin{aligned} \text{(a)} \log_2 0.125 &= \log_2 \left(\frac{1}{8}\right) \\ &= \log_2 (2^{-3}) \\ &= -3 \end{aligned}$$

$$\text{(b)} (e^{\frac{1}{2}x} - e^{-\frac{1}{2}x})^2 = e^x + e^{-x} - 2$$

$$\begin{aligned} \text{(c)} \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) \\ &= \frac{\pi}{4} + \left(-\frac{\pi}{6}\right) \\ &= \frac{\pi}{12} \end{aligned}$$



$$\begin{aligned} \text{(e)} \log_{ab} x &= \frac{\log_a x}{\log_a ab} \\ &= \frac{\log_a x}{\log_a a + \log_a b} \\ &= \frac{\log_a x}{1 + \log_a b} \quad \text{Q.E.D.} \end{aligned}$$

$$\begin{aligned} \text{(f)} \int_0^1 \frac{1}{3+x^2} dx &= \left[\frac{1}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} \right]_0^1 \\ &= \frac{1}{\sqrt{3}} \left(\frac{\pi}{6} - 0 \right) \\ &= \frac{\pi}{6\sqrt{3}} \end{aligned}$$

$$\therefore a = \frac{1}{6\sqrt{3}} = \frac{\sqrt{3}}{18}$$

$$\begin{aligned} \text{(g)} \frac{d}{dx} \ln(\sin^3 x) \\ &= \frac{1}{\sin^3 x} \times 3 \sin^2 x \times \cos x \\ &= 3 \cot x \end{aligned}$$

(h) (i) $-1 \leq x \leq 1$

$$\begin{aligned} \text{(ii)} 2x(1-x) &= 2x - 2x^2 \\ &= -2(x^2 - x) \\ &= -2\left(x^2 - x + \frac{1}{4} - \frac{1}{4}\right) \\ &= -2\left(x - \frac{1}{2}\right)^2 + \frac{1}{2} \end{aligned}$$

$$\therefore \text{Max Value} = \frac{1}{2}$$

(iii) Range:

$$2x - 2x^2 = -1 \text{ for } x = \frac{1}{2} \pm \frac{\sqrt{3}}{2}$$

$$\text{Range: } -\frac{1}{2} \leq y \leq \frac{\pi}{6}$$

Q1 (Cont'd)

$$(i) I = \int_0^{\frac{1}{\sqrt{2}}} \frac{x}{\sqrt{1-x^4}} dx$$

$$u = x^2 \quad \text{When } x=0, u=0$$

$$du = 2x dx \quad x = \frac{1}{\sqrt{2}}, u = \frac{1}{2}$$

$$I = \frac{1}{2} \int_0^{\frac{1}{2}} \frac{du}{\sqrt{1-u^2}}$$

$$= \frac{1}{2} \left[\sin^{-1} u \right]_0^{\frac{1}{2}}$$

$$= \frac{1}{2} \left[\sin^{-1} \frac{1}{2} - \sin^{-1} 0 \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{6} - 0 \right]$$

$$= \frac{\pi}{12}$$

$$(j) y = x^n e^{ax}$$

$$\frac{dy}{dx} = x^n \cdot a e^{ax} + e^{ax} n x^{n-1}$$

$$= e^{ax} (a x^n + n x^{n-1})$$

$$\text{LHS} \Rightarrow \frac{dy}{dx} - ay = n x^{n-1} e^{ax}$$

$$\text{RHS} = \frac{ny}{x}$$

$$= \frac{n x^n e^{ax}}{x}$$

$$= n x^{n-1} e^{ax}$$

$$= \text{LHS}$$

QED

$$(k) \text{ Point A is } \left(\frac{\pi}{4}, 1 \right)$$

$$\text{Line OA: } m = \frac{1}{\frac{\pi}{4}}$$

$$= \frac{4}{\pi}$$

$$\therefore y = \frac{4}{\pi} x$$

Area between chord & line:

$$A = \int_0^{\frac{\pi}{4}} \left(\frac{4}{\pi} x - \tan x \right) dx$$

$$= \int_0^{\frac{\pi}{4}} \left(\frac{4}{\pi} x - \frac{\sin x}{\cos x} \right) dx$$

$$= \left[\frac{2}{\pi} x^2 + \ln(\cos x) \right]_0^{\frac{\pi}{4}}$$

$$= \left(\frac{\pi}{8} + \ln \frac{1}{\sqrt{2}} \right) - (0 + \ln 1)$$

$$= \frac{\pi}{8} - \frac{1}{2} \ln 2$$

$$= \frac{1}{8} (\pi - 4 \ln 2)$$

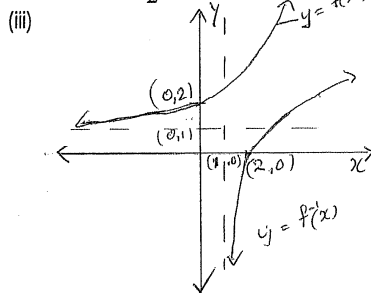
SECTION B

$$\begin{aligned} \text{a)} \quad y &= \frac{1}{\cos x} \\ &= \frac{\cos x \times 0 - 1 \times -\sin x}{\cos^2 x} \quad (\text{quot. rule}) \\ &= \frac{\sin x}{\cos^2 x} \end{aligned}$$

$$\begin{aligned} &= \sec x \tan x \\ \text{b)(i)} \quad f(x) &= 1 + e^{2x} \\ \text{domain: all real } x \\ \text{range: } f(x) &> 1 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \text{let } y &= f(x) = 1 + e^{2x} \\ y - 1 &= e^{2x} \\ \log(y - 1) &= 2x \\ x &= \frac{1}{2} \log(y - 1) \end{aligned}$$

$$f^{-1}(x) = \frac{1}{2} \log(x - 1)$$



$$\text{(iv)} \quad \frac{d}{dx} \left[\frac{1}{2} \log(x - 1) \right] = \frac{1}{2} \left(\frac{1}{x - 1} \right)$$

$$\text{slope of tangent} = \frac{1}{2x - 2}$$

$$\text{slope of normal} = -(2x - 2) = -2 \text{ when } x = 2$$

$$\text{eqn. of normal is } y - 0 = -2(x - 2)$$

$$\text{as } f^{-1}(x) = 0 \text{ when } x = 2$$

$$\text{eqn. of normal is } 2x + y - 4 = 0$$

$$\text{c)} \quad \frac{dy}{dx} = 1 + y, \quad \frac{dx}{dy} = \frac{1}{1 + y}$$

integrating both sides with respect to y

$$x = \ln(1 + y) + c$$

$$0 = \ln 3 + c \text{ as } x = 0 \text{ when } y = 2$$

$$x = \ln(1 + y) - \ln 3$$

$$x = \ln \left(\frac{1 + y}{3} \right)$$

$$e^x = \left(\frac{1 + y}{3} \right)$$

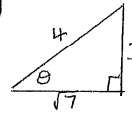
$$y = 3e^x - 1$$

$$\text{(d)} \quad \cos \left[2 \sin^{-1} \left(\frac{3}{4} \right) \right] \text{ let } \theta = \sin^{-1} \left(\frac{3}{4} \right)$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$= 2 \times \frac{7}{16} - 1 \quad (\text{diagram})$$

$$\cos \left[2 \sin^{-1} \left(\frac{3}{4} \right) \right] = -\frac{2}{16} = -0.125$$



(e)

$$\int \frac{3x}{\sqrt{1+x}} dx$$

$$u = 1 + x, \quad x = u - 1$$

$$\frac{du}{dx} = 1, \quad dx = du$$

$$= \int \frac{3u - 3}{\sqrt{u}} du$$

$$= \int (3u^{\frac{1}{2}} - 3u^{-\frac{1}{2}}) du$$

$$= 2u^{\frac{3}{2}} - 6u^{\frac{1}{2}} + c$$

$$= 2\sqrt{(1+x)^3} - 6\sqrt{1+x} + c$$

$$\text{(f)} \quad f(x) = a \cos^{-1} bx, \quad f(0) = 2, \quad f'(0) = 2$$

$$a \cos^{-1}(0) = 2, \quad \cos^{-1}(0) = \frac{\pi}{2}$$

$$\frac{\pi}{2} = \frac{2}{a}, \quad a = \frac{4}{\pi}$$

$$f'(x) = -\frac{ab}{\sqrt{1-b^2x^2}}$$

$$f'(0) = -ab = 2$$

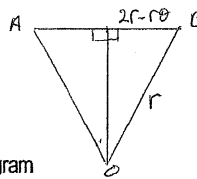
$$-\frac{4}{\pi}b = 2, \quad b = -\frac{\pi}{2}, \quad a = \frac{4}{\pi}$$

(g)(i)

$$\text{perimeter} = AB + \text{arc } AB$$

$$4r = AB + r \times 2\theta$$

$$AB = 4r - 2r\theta$$



$$\sin \theta = \frac{2r - r\theta}{r} \text{ from diagram}$$

$$\sin \theta = 2 - \theta$$

(ii)

$$\sin(1) + 1 - 2 = -0.1585, < 0$$

$$\sin(1.5) + 1.5 - 2 = 0.497, > 0$$

as $f(x)$ is continuous, the sign change means there must be a root between $\theta = 1$ and $\theta = 1.5$.

(iii)

$$f'(\theta) = \cos \theta + 1$$

$$a_1 = 1.25 - \frac{\sin(1.25) + 1.25 - 2}{\cos(1.25) + 1}$$

$$\text{approx. of root} = 1.0987$$

(iv)

$$\theta \approx 1.0987$$

$$2\theta \approx 2.1974^\circ$$

$$2\theta \approx 125^\circ 54'$$

$$\angle AOB = 126^\circ \text{ (nearest degree)}$$

Question (3).

$$(i) v = e^{-2x} (2x^2 - 6)^{\frac{1}{2}}$$

$$\frac{v^2}{2} = \frac{e^{-4x}}{2} (2x^2 - 6)$$

$$= e^{-4x} (x^2 - 3)$$

$$\dot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$a = (2x^2 - 6)(-2e^{-4x}) + \frac{e^{-4x}}{2} (4x)$$

$$a = 2xe^{-4x} - 4e^{-4x}(x^2 - 3)$$

$$a = -2e^{-4x}(2x^2 - x - 6)$$

(ii) max speed

when $\dot{x} = 0$

$$\text{Now, } -2e^{-4x} < 0, \forall x$$

$$\therefore 2x^2 - x - 6 = 0$$

$$(2x+3)(x-2) = 0$$

$$\therefore x = -\frac{3}{2}, 2$$

$$\therefore x \geq 1.8 \quad \therefore x = 2$$

$$(iii) v = \frac{dx}{dt} = \frac{-2x}{\sqrt{2x^2 - 6}}$$

$$\therefore \frac{dt}{dx} = \frac{e^{2x}}{\sqrt{2x^2 - 6}}$$

$$T = \int_2^3 \frac{e^{2x}}{\sqrt{2x^2 - 6}} dx$$

$$(iv) \int_2^3 \frac{e^{2x}}{\sqrt{2x^2 - 6}} dx$$

$$= \frac{h}{3} [(y_1 + y_3) + 4y_2]$$

$$h = \frac{1}{2} \left[\frac{1}{6} [(y_1 + y_3) + 4y_2] \right]$$

$$= \frac{1}{6} \left[\frac{e^4}{\sqrt{2}} + \frac{e^6}{2\sqrt{3}} + 4 \left(\frac{e^5}{\sqrt{6.5}} \right) \right]$$

$$= \frac{1}{6} [38.61 + 116.46 + 232.85]$$

$$= 64.65$$

$$(b) v = \pi \int_0^R x^2 dy$$

$$(i) x^4 = 16y, \quad x^2 = 4y^{\frac{1}{2}}$$

$$\therefore v = 4\pi \int_0^R y^{\frac{1}{2}} dy$$

$$= 4\pi \left[\frac{2}{3} y^{\frac{3}{2}} \right]_0^R$$

$$= \frac{8\pi}{3} h^{\frac{3}{2}}$$

[24 marks]

(ii)

$$\frac{dv}{dt} \propto \sqrt{R}$$

$$\therefore \frac{dv}{dt} = -k\sqrt{R} \quad (1)$$

where k is a constant

$$\text{Now, } \frac{dR}{dt} = \left(\frac{dv}{dR} \right) \times \frac{dv}{dt} \quad (1)$$

$$= \frac{1}{4\pi\sqrt{R}} \times -k\sqrt{R}$$

$$= -\frac{k}{4\pi} \quad (\text{a constant})$$

(c)

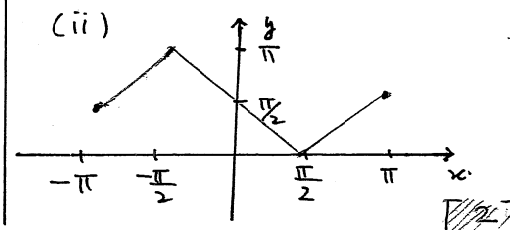
$$(i) y = \cos^{-1}(\sin x)$$

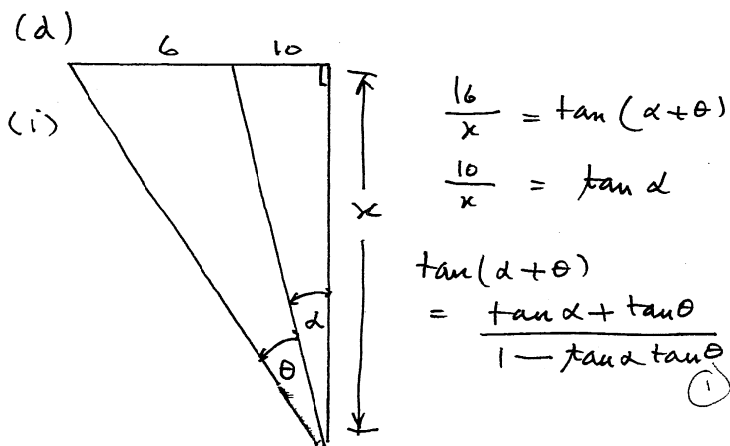
$$\frac{dy}{dx} = \frac{-\cos x}{\sqrt{1 - \sin^2 x}}$$

$$= \frac{-\cos x}{|\cos x|}$$

$$= \pm 1$$

(ii)





$$\therefore \frac{6}{x} = \frac{\frac{10}{x} + \tan \theta}{1 - \frac{10}{x} \tan \theta}$$

$$= \frac{\left(\frac{10 + x \tan \theta}{x} \right)}{\left(\frac{x - 10 \tan \theta}{x} \right)} \quad (1)$$

$$10x + x^2 \tan \theta = 16x - 160 \tan \theta$$

$$(160 + x^2) \tan \theta = 6x$$

$$\therefore \tan \theta = \frac{6x}{x^2 + 160}$$

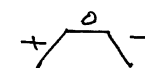
(ii) Let $E = \tan \theta$.

(ii). $\frac{dE}{dx} = \frac{6(160 + x^2) - 6x(2x)}{(160 + x^2)^2} \quad (1)$

$$= \frac{6(160 - x^2)}{(160 + x^2)^2}$$

$$\frac{dE}{dx} = 0, \Rightarrow x^2 = 160, x = \sqrt{160}$$

x	12	$4\sqrt{10}$	$4\sqrt{11}$
$\frac{dE}{dx}$	+	0	-



$$\therefore x = \sqrt{160}$$

(1) for maximum E

(iii)

$$\tan \theta = \frac{6(4\sqrt{10})}{160 + 160} \quad (1)$$

$$= \frac{24\sqrt{10}}{320}$$

$$= \frac{3\sqrt{10}}{40}$$

$$= \frac{3\sqrt{10}}{\sqrt{1600}} = \frac{3}{\sqrt{160}}$$

$$\therefore \theta = \tan^{-1}\left(\frac{3}{\sqrt{160}}\right)$$